

Medical Robotics

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Lecture 8

Virtual Fixtures

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1. Introduction to Virtual Fixtures

1.1. Definition and Historical Origins

The concept of **virtual fixtures** (VFs) was introduced in 1992 by L. B. Rosenberg at the US Air Force Armstrong Laboratory, in the context of telerobotic manipulation. Rosenberg's original motivation was to study:

“the beneficial effects of corrupting the link between operator and remote environment by introducing abstract perceptual information into the interface called virtual fixtures.”

A virtual fixture is not a physical object. It is a software-generated constraint on the motion of a robot manipulator (or on the haptic feedback presented to a teleoperator) that mimics the effect of a physical fixture or guide. The analogy often used is that of a **ruler guiding a pencil**: the ruler does not draw the line itself, but it constrains the pencil to move along a preferred direction and prevents unintended deviation.

Virtual fixtures offer an **alternative to direct force feedback from the environment**: rather than reflecting the actual contact forces between the tool and tissue, they provide synthetic sensory cues that improve the safety, accuracy, and speed of robot-assisted manipulation tasks. They can be applied to both **cooperative (hands-on) manipulation** and **telemanipulation** systems, making them one of the most broadly applicable concepts in medical robotics control.

Relationship to active constraints. The Acrobot's active constraint controller (Lecture 7) is a specific example of a virtual fixture applied in a cooperative manipulation context. The three-region partition, the admittance-based filtering of the surgeon's force, and the stiffening near the boundary are all implementations of a *Forbidden-Region Virtual Fixture* (FRVF). The virtual fixture framework introduced in this lecture generalises and formalises these ideas.

1.2. Placement in the Medical Robotics Landscape

The following table, taken from the course slides, summarises the control modalities and sensor/actuator characteristics across the main domains of medical robotics. Virtual fixtures are relevant in every row of this table, as they can be layered onto cooperative, teleoperated, or semi-autonomous controllers.

Domain	Robotic Task	Kinematic tecture	Archi- ture	Control Modality	Sensors/Actuators
Orthopedics	Machining of rigid surface	Conventional, 5–6 dof	5–6	Autonomous, cooperative, hands-on	Conventional; vel: mm/s; force: up to 100 N
MIS	Constrained manipulation	RCM, 5–6 external + extra internal dof	external + internal dof	Teleoperation	Conventional; vel: cm/s; force: few N
Neurosurgery, radiology, radiotherapy	Constrained targeting	Conventional dedicated end	+ front-	Teleoperation, semi-autonomous	MR/CT compatible (pneumatic, ultrasonic); force: few N
Microsurgery	Micromanipulation	Dedicated kinematic architecture	kinematic architecture	Shared/cooperative teleoperation	Piezo/ultrasonic; force: few mN
Tele-echo, TMS, skin harvesting	Surface tracking	Conventional dedicated wrist	+ wrist	Autonomous, teleoperation	Conventional; vel: mm/s; force: few N

2. Types of Virtual Fixtures

There are two principal categories of virtual fixtures:

Guidance Virtual Fixtures (GVFs). A GVF assists the user in moving the manipulator *along* desired paths or surfaces in the workspace. The fixture defines a *preferred direction* of motion; forces (or velocities) in the preferred direction are transmitted with little or no attenuation, while those in non-preferred directions are filtered or resisted.

Forbidden-Region Virtual Fixtures (FRVFs). An FRVF prevents the manipulator from *entering* a defined forbidden region of the workspace. It acts as a virtual wall: motion away from or along the boundary is permitted, while motion into the forbidden region is blocked or strongly resisted.

Both GVFs and FRVFs can be implemented using either an **admittance-type** or an **impedance-type** force/motion relationship. As discussed in the lectures on impedance/admittance control:

- **Admittance-type VF:** $V(s) = Y(s)F(s)$ — force is the input, velocity (or position) is the output.
- **Impedance-type VF:** $F(s) = Z(s)V(s)$ — velocity (or displacement) is the input, force is the output.

3. Admittance-Type Virtual Fixtures

3.1. Guidance VFs

In an admittance-type GVF, the admittance matrix $Y(s)$ is chosen to be **anisotropic**: different in different Cartesian directions. By attenuating the applied force components in the non-preferred directions, a passive guidance is created along the preferred direction of motion.

The level of guidance is characterised by the response in the non-preferred directions:

- **Hard guidance:** none (or almost none) of the non-preferred force component is transmitted into motion. The user has no or very little freedom to deviate from the preferred path.
- **Soft guidance:** the non-preferred force component is attenuated but not eliminated, giving the user some freedom to deviate from the path.

Experiments in microsurgical environments with admittance-controlled cooperative robots have shown that operators perform best in uncertain environments with a **medium-level GVF**, because the virtual fixture planner is never perfect. Hard guidance can trap the operator if the planned path is slightly wrong; soft guidance gives the flexibility to correct.

3.2. Forbidden-Region VFs

Admittance-type FRVFs are, in effect, a subclass of GVFs for admittance-controlled cooperative robots. The Acrobot's active constraint controller (Lecture 7) is the canonical example: the three-region partition and the direction-dependent admittance constitute an admittance-type FRVF.

In **telemanipulation**, the admittance-type FRVF is concerned principally with preventing the *slave* manipulator from penetrating the forbidden region. Penetration of the master device into the corresponding region of its workspace has no direct physical consequences, though haptic feedback can be generated for the operator based on the depth of penetration.

4. Impedance-Type Virtual Fixtures

4.1. Guidance VFs

An impedance-type GVF acts as a **potential field** that actively influences the motion of the robotic manipulator. The impedance matrix $\mathbf{Z}(s)$ relates the velocity (or displacement) of the manipulator to a force that the controller applies. As with the admittance case, $\mathbf{Z}(s)$ is chosen anisotropically to steer or constrain motion.

4.2. Forbidden-Region VFs

Impedance-type FRVFs can be implemented on telemanipulators by **overlaying a penalty-based virtual wall** on the existing telemannipulation controller. The virtual wall can be placed on the master side, the slave side, or both:

- On the **master side**: the operator feels a resistive force from the haptic device when approaching the boundary. This directly communicates the constraint to the operator but does not guarantee that the slave will not violate it (if the operator forces through).
- On the **slave side**: the slave robot's controller resists entry into the forbidden region regardless of the master command. This guarantees physical safety but the operator may not feel the constraint.
- **Both sides**: combines the safety guarantee of the slave-side constraint with the informative haptic feedback of the master-side constraint.

Each placement offers a different haptic experience for the user, depending on the underlying telemannipulation controller.

Torque control requirement. As emphasised in the lecture, impedance-type control requires **torque control** capability at the robot joints. Many industrial robots expose only position-based interfaces (no direct torque access). In those cases, only admittance-type control is feasible. This practical constraint often drives the choice between the two

paradigms when deploying virtual fixtures on commercially available robotic platforms.

5. Example 1: Guidance of a Point Mass

5.1. System Setup

To illustrate the admittance and impedance VF control laws concretely, consider a simplified planar system: a mass m that translates in the plane under the action of a control force $\mathbf{U}$ and a guiding force $\mathbf{F}_G$ (the force applied by the operator/surgeon), with no friction. The state is:

$$\mathbf{x} = (x_1, x_2)^T, \quad \mathbf{v} = (\dot{x}_1, \dot{x}_2)^T, \quad m\dot{\mathbf{v}} = \mathbf{U} + \mathbf{F}_G$$

5.2. Admittance-Based Control

From an admittance-type GVF, the desired velocity of the mass is set by the admittance relation:

$$\mathbf{V}(s) = \mathbf{Y}(s)\mathbf{F}_G(s)$$

Choosing $\mathbf{V}_d(s) = \mathbf{V}(s)$ and applying the control law:

$$\mathbf{U} = m\dot{\mathbf{v}}_d + \mathbf{K}_D(\mathbf{v}_d - \mathbf{v}) + \mathbf{K}_P(\mathbf{x}_d - \mathbf{x})$$

makes the system track the desired velocity, so it moves according to the designed admittance $\mathbf{Y}(s)$.

For a **soft guidance** fixture, the admittance matrix takes the diagonal form:

$$\begin{pmatrix} \dot{x}_d \\ \dot{y}_d \end{pmatrix} = \begin{pmatrix} k_{a,x} & 0 \\ 0 & k_{a,y} \end{pmatrix} \begin{pmatrix} F_{G,x} \\ F_{G,y} \end{pmatrix} = \mathbf{Y} \begin{pmatrix} F_{G,x} \\ F_{G,y} \end{pmatrix}$$

where $k_{a,x} \neq k_{a,y}$ creates the directional asymmetry (the preferred direction has the higher admittance). The control input is:

$$\begin{pmatrix} U_x \\ U_y \end{pmatrix} = m \begin{pmatrix} \ddot{x}_d \\ \ddot{y}_d \end{pmatrix} + \begin{pmatrix} k_{v,x} & 0 \\ 0 & k_{v,y} \end{pmatrix} \begin{pmatrix} \dot{x}_d - \dot{x} \\ \dot{y}_d - \dot{y} \end{pmatrix} + \begin{pmatrix} k_{p,x} & 0 \\ 0 & k_{p,y} \end{pmatrix} \begin{pmatrix} x_d - x \\ y_d - y \end{pmatrix}$$

For **hard guidance**, $k_{a,y} \rightarrow 0$ (or is set to zero), completely suppressing motion in the non-preferred y -direction.

This control scheme has an instructive **equivalent formulation** in the position domain with dynamic impedance (see slides on interaction control). Define the deviation of the reference position from the nominal desired position as $\Delta p_r = p_r - p_d$. The admittance model produces:

$$\Delta p_r = \frac{F_{\text{ext}}}{ms^2 + bs + k}$$

which is the response of a mass-spring-damper to the external force. The block diagram (see slides) shows a position controller (inner loop) that tracks $p_r = p_d + \Delta p_r$, where Δp_r is generated by the admittance model from the external force F_{ext} .

5.3. Impedance-Based Control

In the impedance-based approach, the control $\mathbf{U}$ is designed so that the closed-loop dynamics of the interaction matches the behaviour of an **impedance model**:

$$\mathbf{F}_G(s) = \mathbf{Z}(s)\mathbf{V}(s).$$

Concretely, specifying a desired apparent mass m_d , damping $\mathbf{K}_D > 0$, and stiffness $\mathbf{K}_P > 0$ with respect to a motion reference $\mathbf{x}_d$ gives the desired interaction law:

$$m_d(\dot{\mathbf{v}} - \dot{\mathbf{v}}_d) + \mathbf{K}_D(\mathbf{v} - \mathbf{v}_d) + \mathbf{K}_P(\mathbf{x} - \mathbf{x}_d) = \mathbf{F}_G$$

This is achieved by the control law:

$$\mathbf{U} = \frac{m}{m_d}(m_d\dot{\mathbf{v}}_d + \mathbf{K}_D(\mathbf{v}_d - \mathbf{v}) + \mathbf{K}_P(\mathbf{x}_d - \mathbf{x})) + \left(\frac{m}{m_d} - 1\right)\mathbf{F}_G$$

When $m_d = m$ (compliance control), the last term vanishes and **no measurement of $\mathbf{F}_G$ is needed**, which is a practically important simplification.

Case study (see slides): the reference task for the impedance-based guidance is a sinusoidal trajectory:

$$\begin{pmatrix} x_d(t) \\ y_d(t) \end{pmatrix} = \begin{pmatrix} A_x \sin(\omega_x t) \\ A_y \sin(\omega_y t) \end{pmatrix}$$

The stiffness matrices $\mathbf{K}_{P,I}$ are again made diagonal and anisotropic to produce hard or soft guidance: high stiffness in the non-preferred direction gives hard guidance (the mass closely tracks the task trajectory even under external perturbation), while low stiffness gives soft guidance (the operator can displace the mass from the trajectory, but the stiffness acts as a restoring spring that returns the mass to the task after the interaction ends).

5.4. Comparison of Admittance and Impedance Schemes

Both the admittance-based and the impedance-based VF controllers lead to the **same closed-loop interaction dynamics** (for the simplified mass system):

$$m(\dot{\mathbf{v}} - \dot{\mathbf{v}}_d) + \mathbf{K}_D(\mathbf{v} - \mathbf{v}_d) + \mathbf{K}_P(\mathbf{x} - \mathbf{x}_d) = \mathbf{F}_G$$

They differ in how this target behavior is achieved:

Admittance-based	Impedance-based
$\mathbf{V}(s) = \mathbf{Y}(s)\mathbf{F}_G(s)$: force $\rightarrow$ velocity reference	$\mathbf{F}_G(s) = \mathbf{Z}(s)\mathbf{V}(s)$: velocity/position $\rightarrow$ force
Outer admittance model generates $\mathbf{V}_d$; inner low-level controller tracks $\mathbf{V}_d$	Impedance model directly generates control $\mathbf{U}$
Suitable for <i>position-controlled</i> robots (no torque access needed at the outer level)	Requires <i>torque control</i> capability
Force measurement ($\mathbf{F}_G$) needed for the admittance model	Force measurement can be avoided when $m_d = m$

Key practical distinction. As discussed in the lecture: if the robot platform only exposes a position control interface and does not allow direct joint torque commands, impedance-based control cannot be implemented. In that case, admittance control is the only viable option. This is a common situation with industrial robots re-purposed for medical applications.

6. Student Project Examples

The following examples, developed by students of this course, concretely illustrate the application of the VF framework in different contexts. They are cited here both as pedagogical illustrations and as a reminder that the concepts of this course can be implemented on accessible hardware.

6.1. Forbidden-Region VF for Rehabilitation

A haptic interface (3D Systems Touch) was used to implement a **forbidden-region VF for wrist rehabilitation**. The task was framed as a maze navigation game: the operator moves a virtual ball through a maze displayed on a screen, and the haptic device generates resistive forces when the ball approaches or contacts the maze walls. The FRVF prevents penetration of the virtual walls, delivering force feedback that the patient experiences as the walls of the maze. This gamification approach is used in commercial rehabilitation systems — particularly for paediatric patients — because it increases engagement and makes therapy feel less like a clinical exercise.

6.2. Guidance VF for Needle Insertion

A guidance VF was implemented to assist **robotic needle insertion** into a simulated patient surface. The scenario used a haptic master interface connected to a virtual KUKA robot with a needle at the end-effector. The task requires inserting the needle *perpendicularly* to the patient surface.

Without the VF, the operator must carefully orient the master device to achieve a perpendicular approach, which is ergonomically demanding. With the GVF, the fixture filters the master motion and *only transmits the component that drives the needle perpendicularly* to the surface. The remaining components (lateral translations) are suppressed. As a result, the reference sent to the slave robot corresponds to a corrected, perpendicular insertion trajectory, regardless of how the operator moves the master. This is a direct application of the hard guidance concept to a clinically motivated task.

Depth-camera-based FRVF. An additional example shown in the lecture demonstrated a real-time FRVF system where the forbidden regions are defined dynamically from **depth camera** data (point cloud). The operator moves a haptic device on the master side and feels opposing forces when the slave end-effector approaches an object identified in the point cloud. A hard FRVF prevents any contact with the protected object; a soft FRVF allows contact only if the operator applies sufficient force to overcome the virtual wall. This real-time, sensor-driven approach generalises the concept beyond pre-planned geometric constraints.

7. Summary and Key Conceptual Threads

- **Virtual fixtures generalise active constraints.** The Acrobot's active constraint controller is a special case of a forbidden-region VF. The VF framework provides a unified vocabulary (GVF/FRVF, admittance/impedance, hard/soft) that applies across cooperative manipulation and telemanipulation contexts.
- **Admittance and impedance are dual implementations of the same interaction law.** Both lead to a mass-spring-damper closed-loop interaction between the robot and the operator. The choice between them depends available control input and sensors.
- **The anisotropy of $Y(s)$ or $Z(s)$ is the mechanism of guidance.** Setting different gains in different Cartesian directions is all that is needed to produce directional guidance or constraint enforcement.
- **Hard vs. soft guidance is a design trade-off.** Hard guidance maximises task accuracy but reduces the operator's ability to correct for planning errors. Soft guidance sacrifices some accuracy for flexibility. Medium-level guidance is empirically the best compromise in uncertain surgical environments.

- **Virtual fixtures are not limited to pre-planned geometry.** Real-time VF generation from depth sensors or intra-operative imaging extends the concept to environments where the geometry of forbidden regions is not fully known pre-operatively.

Key topics for revision.

1. The definition of a virtual fixture, its historical origin (Rosenberg, 1992), and the analogy with a physical fixture.
2. The two main types: GVF and FRVF, and how each type improves safety, accuracy, or speed.
3. Admittance-type VFs: $\mathbf{V}(s) = \mathbf{Y}(s)\mathbf{F}_G(s)$, the role of anisotropy in creating guidance, and hard vs. soft guidance.
4. Impedance-type VFs: $\mathbf{F}_G(s) = \mathbf{Z}(s)\mathbf{V}(s)$, the desired interaction law, the control law for the point mass, and when force measurement can be omitted ($m_d = m$).
5. The admittance-based control law for the planar mass (the matrices $\mathbf{Y}$, $\mathbf{K}_D$, $\mathbf{K}_P$ and how they relate to hard/soft guidance).
6. The comparison between admittance and impedance: same interaction dynamics, different implementation, different requirements on robot hardware.
7. The practical constraint that impedance control requires torque control, while admittance control can be implemented on position-controlled robots.
8. Placement options for slave-side vs. master-side FRVFs in telemanipulation and their different haptic consequences.

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